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Morphometric Analysis of Lumbar Intervertebral Disc Height: An Imaging Study

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■ BACKGROUND: **Little published data exist regarding normal values of disc height. Current literature relies on plain radiographs making accurate measurements of individual lumbar disc height difficult.**

■ OBJECTIVE: **We seek to establish normal values for lumbar intervertebral discs in different age groups using computed tomography scans in healthy individuals.**

■ METHODS: **Two hundred forty anonymized abdominal computed tomography scans (131 women) were prospectively collected once institutional review board approval was obtained. Individuals with spinal pathologies were excluded. Disc height measurements were obtained at the anterior edge, center, and posterior edge of each vertebra in the midsagittal plane, averaged, and compared against age and sex.**

■ RESULTS: **Average age was 45 (14–83) years for women and 48 (14–89) years for men. Average lumbar disc height was 5.6 ± 1.1 mm for men and 4.8 ± 0.8 mm for women at T12/L1, 6.9 ± 1.3 mm for men and 5.8 ± 0.9 mm for women at L1/2, 8.1 ± 1.4 mm for men and 6.9 ± 1.1 mm for women at L2/3, 8.7 ± 1.5 mm for men and 7.6 ± 1.2 mm for women at L3/4, 9.2 ± 1.6 mm for men and 8.5 ± 1.6 mm for women at L4/5, and 8.8 ± 1.6 mm for men and 8.6 ± 1.8 mm for women at L5/S1. Disc height was significantly smaller for women than men ($P < 0.001$), except at L5/S1.**

■ CONCLUSIONS: **Variation in disc height is determined much more by sex than age. The maximum height of the interbody space in the adult lumbar spine was at the L4/5 level (8.9 ± 1.7 mm [men], 8.6 ± 1.8 mm [women]). Based on our findings, >10 mm cage height will result in supraphysiologic interbody space restoration and potentially predispose to complications.**

INTRODUCTION

The main goals of spine surgery include direct or indirect decompression of neural elements, restoration of disc height, sagittal and coronal balance, and solid fusion when indicated. Interbody fusion using an intervertebral spacer (cage) has become one of the most widely used methods for achieving these goals. However, oversized cage selection can result in increased distractive and compressive forces on endplates,[1-4] which can predispose the patient to endplate subsidence, vertebral body fracture, loss of correction, and failure of the procedure with recurrence of symptomatic pathology and the need for complex revision surgery.[5-7] Inappropriate cage selection can potentially predispose the patient to unnecessary future complications, and the most anatomically correct cage size should be used to avoid any such potential problems. Published data regarding normal values of disc height in the aging spine are limited. Current literature that relies on plain radiographs requires complex corrections, making accurate measurement of individual lumbar disc height difficult.[8,9] Magnetic resonance imaging data, although more easily interpreted, still relies on 2 dimensional sagittal section, and is not consistently subcategorized by lumbar level nor by decade of life.[10-13] Existing data from computed tomography (CT) scans are similarly limited and also scarce.[14] We seek to establish normal values for lumbar intervertebral discs using novel CT scan-based vertebral modeling in healthy individuals to provide a guide in the selection of interbody prosthetics.

Key words
- Intervertebral disc
- Intervertebral disc degeneration
- Lumbar intervertebral disc height
- Lumbar vertebrae
- Spinal fusion
- Spine

Abbreviations and Acronyms
AP: Anterior-posterior
BMI: Body mass index
CT: Computed tomography
FH: Foraminal height

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MATERIALS AND METHODS

After the University of South Florida institutional review board approval, 240 anonymized abdominal CT scans (131 women and 109 men) were prospectively collected at the University of South Florida from January 2010 and August 2012. Data were collected in individuals aged 10–80 years who presented with a non-spinal complaint in whom thoracic and lumbar CT was obtained. Patients were scanned for any spinal pathology by analyzing individual CTs. Individuals with incidental spinal pathologies were excluded. A dedicated neuroradiologist blinded to patient age and sex reviewed each CT and performed individual measurements. The CT data were analyzed using Mimics Innovation Suite software version 19.0 (Materialise, Leuven, Belgium). Spinal vertebrae were modeled with threshold conditions of 226-3071 Hounsfield units. Hand segmentation tools were used to isolate each individual vertebral body by separating the superior and inferior articular processes from the adjacent vertebral body (and 12th rib for T12). Disc height measurements were obtained at the anterior edge, center, and posterior edge of each vertebra in the midsagittal plane (Figure 1). These were also averaged as a measure of overall disc height. A second disc height average was also calculated using only the anterior and posterior values. Left and right neuroforaminal heights were measured at the distances furthest apart along the superior and inferior pedicle borders.

All measurements were averaged and compared against age and sex. Independent *t*-tests were run comparing each measurement value based on sex. A paired *t*-test was used to compare left and right foraminal height (FH). The Pearson correlation coefficient was calculated for each measurement testing for age correlation

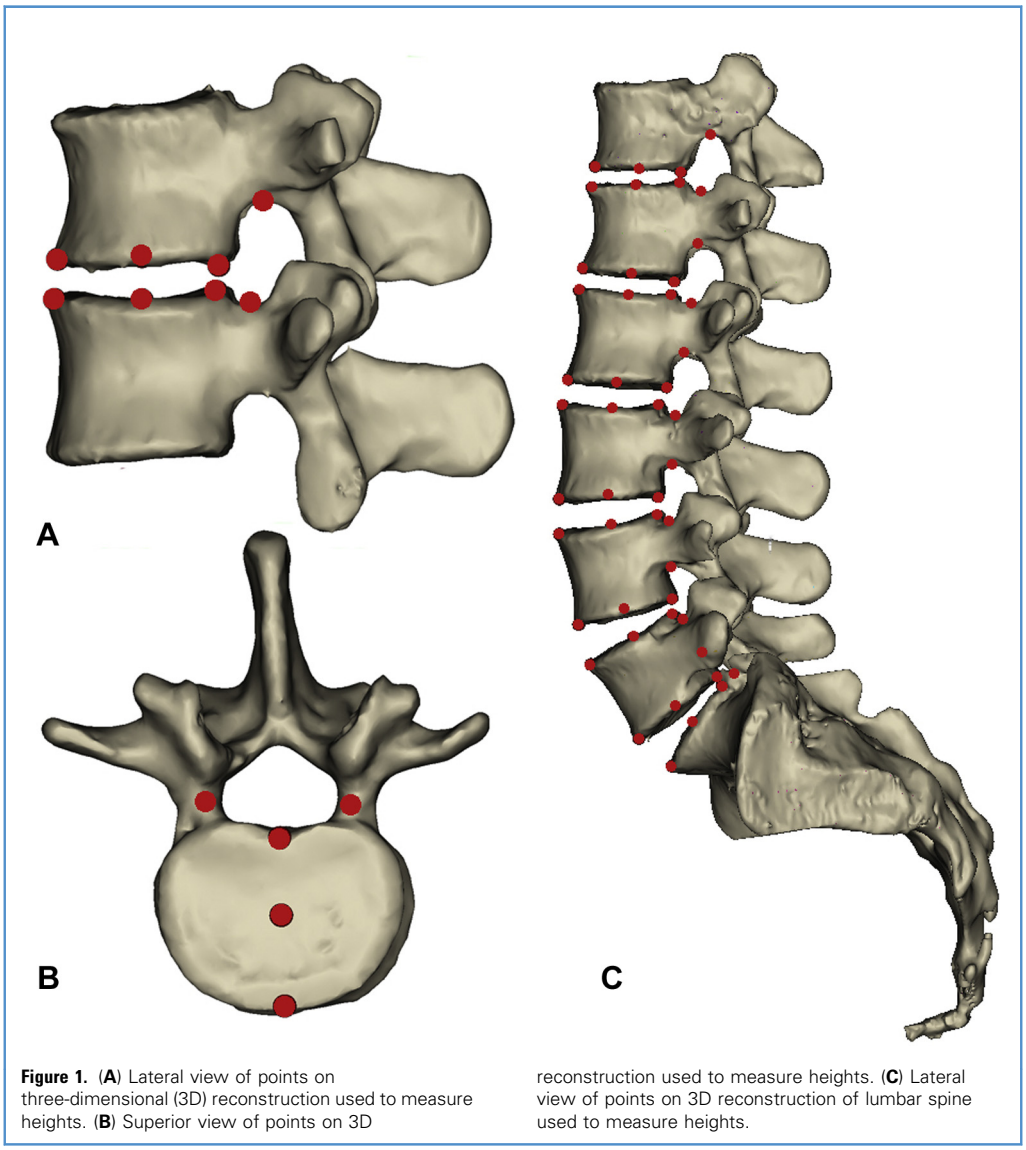


Figure 1. (**A**) Lateral view of points on three-dimensional (3D) reconstruction used to measure heights. (**B**) Superior view of points on 3D reconstruction used to measure heights. (**C**) Lateral view of points on 3D reconstruction of lumbar spine used to measure heights.

Table 1. Patient Volume per Decade of Life

Decade	Number Women	Number Men
10	8	8
20	25	19
30	18	17
40	23	8
50	25	23
60	17	22
70	15	12
Total	131	109

with corresponding strength as described in literature.[15] Height, weight, and body mass index (BMI) data were collected when available. Independent sample t-test was used to compare male subject to female subject height, weight, and BMI. The Pearson correlation was run to determine if there was a correlation between age and height, weight, or BMI. All statistical tests were run in SPSS software package, version 24.0 (IBM Corp., Armonk, New York, USA) with a P value of 0.05 being considered significant.

RESULTS

Average age was 45 (14–79) years for women and 47 (14–79) years for men. Table 1 shows the number of men and women included in the study for each decade of life. Figures 2–7 graphically illustrate variation in height for each level. Detailed anterior, posterior, middle, and average disc heights are presented in Table 2 for men and Table 3 for women. Detailed FH is presented in Table 4 for men and Table 5 for women. Average lumbar disc height was 5.6 ± 1.1 mm for men and 4.8 ± 0.8 mm for women at T12/L1, 6.9 ± 1.3 mm for men and 5.8 ± 0.9 mm for women at L1/2, 8.1 ± 1.4 mm for men and 6.9 ± 1.1 mm for women at L2/3, 8.7 ± 1.5 mm for men and 7.6 ± 1.2 mm for women at L3/4, 9.2 ± 1.6 mm for men and 8.5 ± 1.6 mm for women at L4/5, and 8.8 ± 1.6 mm for men and 8.6 ± 1.8 mm for women at L5/S1 (Figure 8).

Disc height was significantly smaller for women than men ($P < 0.001$) except L5/S1, in which it was approximately the same height. FH was also significantly smaller for women than for men

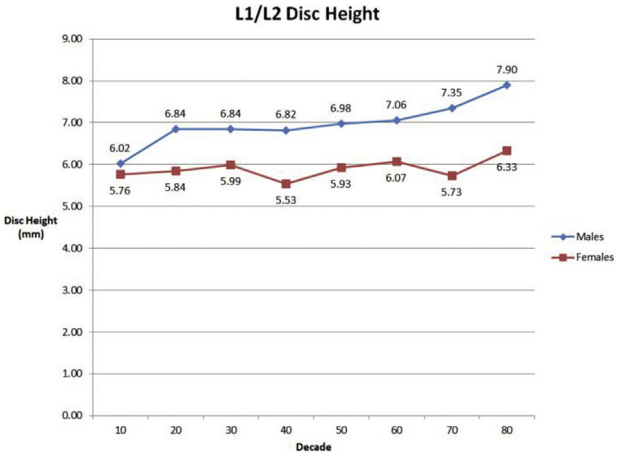


Figure 3. Variation with age of average disc height and anterior-posterior average disc height by decade for male subjects and female subjects at L1/2.

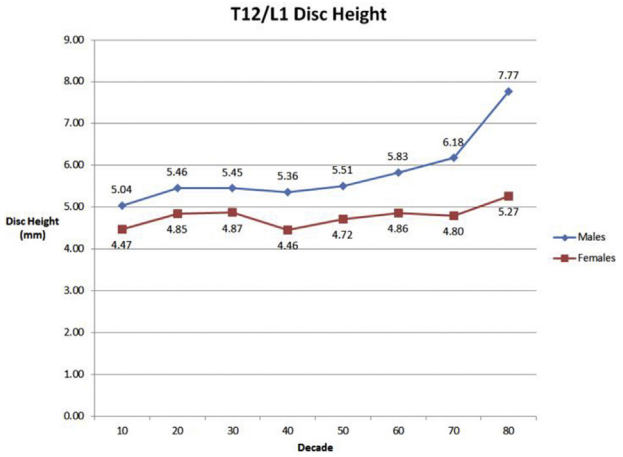


Figure 2. Variation with age of average disc height and anterior-posterior average disc height by decade for male subjects and female subjects at T12/L1.

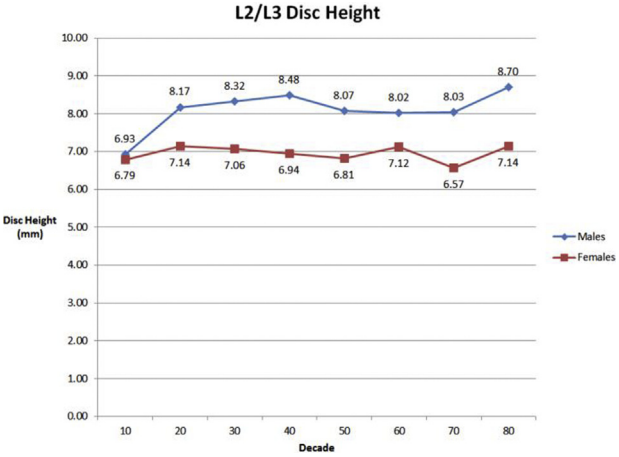


Figure 4. Variation with age of average disc height and anterior-posterior average disc height by decade for male subjects and female subjects at L2/3.

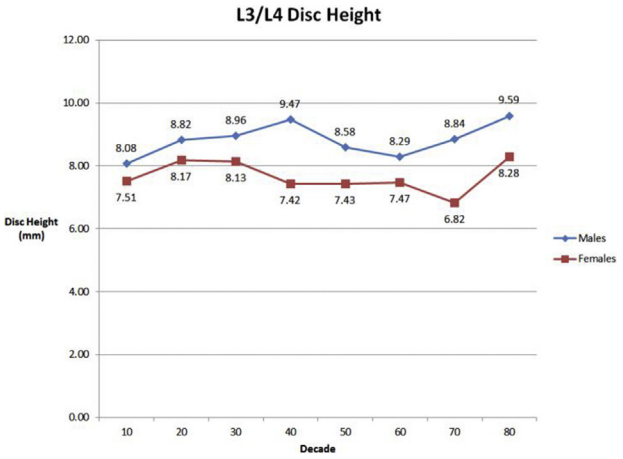


Figure 5. Variation with age of average disc height and anterior-posterior average disc height by decade for male subjects and female subjects at L3/4.

for all levels except L4/5 and L5/S1. (see **Table 6** for independent t-test for disc and FH).

The Pearson correlation was used to determine relationship between age by decade for both disc height and of FH. In men, T12/L1 middle disc height showed a significant moderate positive correlation to age, and posterior and average disc height showed a weak positive significant correlation to age. At L1/2, middle and average disc height showed a significant weak positive correlation to age. At L2/3, middle disc height showed a weak positive correlation to age. L3/4 did not demonstrate any significant correlation for any disc height measurements to age. L4/5 anterior, posterior, and anterior-posterior (AP) average disc heights demonstrated a weak negative significant correlation with age. L5/S1 did not demonstrate any significant correlation for any disc height measurements to age. FH demonstrated a weak or moderate negative significant correlation with age for all levels, except T12/L1 and L1/2. Complete results of the Pearson correlation of disc height and FH measurements with age can be found in **Table 7**.

For women, T12/L1 disc did not demonstrate any significant correlation. L1/2 showed a significant weak positive correlation for middle disc height, and a significant weak negative correlation for posterior disc height. L2/3 showed a significant weak negative correlation to age for posterior disc height, and AP disc height average. L3/4 and L4/5 showed a significant moderate negative correlation between posterior disc height and age, and a significant weak negative correlation for anterior disc height, AP average height, and average disc height with age. L5/S1 showed a significant weak negative correlation for anterior disc height, AP average height, and average disc height with age. FH at all levels showed a significant moderate to strong negative correlation to age. Complete detailed Pearson correlation results for women are shown in **Table 8**.

Height, weight, and BMI data were available for 81 female subjects and 69 male subjects. Mean height and weight were significantly larger in men than in women ($P < 0.0005$ in both instances). Complete age, height, weight, and BMI data can be seen in **Table 9** for women and **Table 10** for men. Statistical comparisons of male subject and female subject height, weight, and BMI are shown in **Table 11**. No correlation between height, weight, or BMI and age were found, except that height negatively correlated to age in women ($R = -0.332$, $P = 0.002$). Complete correlation results are shown in **Table 12**.

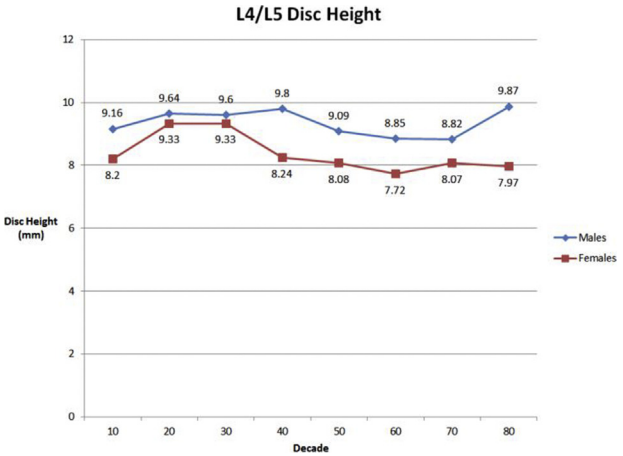


Figure 6. Variation with age of average disc height and anterior-posterior average disc height by decade for male subjects and female subjects at L4/5.

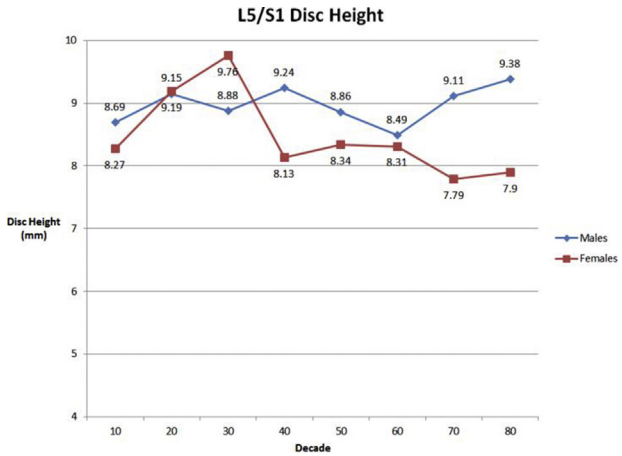


Figure 7. Variation with age of average disc height and anterior-posterior average disc height by decade for male subjects and female subjects at L5/S1.

Table 2. Male Disc Height by Decade of Life

Level	Location	First Decade	Second Decade	Third Decade	Fourth Decade	Fifth Decade	Sixth Decade	Seventh Decade
T12/L1	Anterior	5.38 ± 1.6	6.7 ± 1.6	6.24 ± 1.7	5.7 ± 1.6	6.15 ± 1.4	6.39 ± 1.3	6.78 ± 1.4
	Middle	5.63 ± 1.9	5.84 ± 1	6.47 ± 1.2	7.15 ± 1.2	6.78 ± 1	7.14 ± 1.6	7.48 ± 1
	Posterior	3.43 ± 1.1	3.83 ± 1.1	3.57 ± 0.8	3.57 ± 0.9	3.63 ± 1	3.95 ± 0.9	4.28 ± 1.1
	Average	4.72 ± 1.4	5.46 ± 1.1	5.43 ± 1	5.47 ± 1.2	5.52 ± 0.9	5.83 ± 1.1	6.18 ± 1.1
	AP Avg	4.26 ± 1.3	5.27 ± 1.3	4.91 ± 1	4.63 ± 1.2	4.89 ± 1	5.17 ± 1	5.53 ± 1.2
L1/2	Anterior	6.57 ± 2.1	8.26 ± 1.8	7.55 ± 1.3	7.18 ± 0.9	7.67 ± 1.3	7.41 ± 1.5	7.75 ± 2.1
	Middle	6.83 ± 2.4	7.35 ± 1	7.85 ± 1.1	8.89 ± 1.1	8.44 ± 1.4	8.81 ± 2.6	8.9 ± 1.6
	Posterior	4.12 ± 1.7	4.91 ± 1.2	5.01 ± 1.1	4.85 ± 0.9	5.1 ± 1.4	4.96 ± 1.3	5.4 ± 1.1
	Average	5.84 ± 2	6.84 ± 1.2	6.8 ± 0.9	6.97 ± 0.8	7.06 ± 1.1	7.06 ± 1.7	7.35 ± 1.4
	AP Avg	5.35 ± 1.85	6.58 ± 1.3	6.28 ± 1	6.02 ± 0.8	6.37 ± 1.2	6.19 ± 1.3	6.58 ± 1.4
L2/3	Anterior	7.23 ± 1.7	9.48 ± 1.8	9.12 ± 1.5	8.34 ± 0.8	8.78 ± 1.7	8.46 ± 1.4	8.38 ± 1.9
	Middle	8.01 ± 1.7	9.17 ± 1.4	9.84 ± 1.1	11.02 ± 1.5	9.85 ± 1.8	10.24 ± 2.5	10.11 ± 1.7
	Posterior	4.6 ± 1.3	5.85 ± 1.3	6.02 ± 1.1	6.4 ± 1.2	5.68 ± 1.7	5.38 ± 1.5	5.61 ± 1.6
	Average	6.61 ± 1.5	8.17 ± 1.3	8.32 ± 1	8.59 ± 1	8.1 ± 1.5	8.02 ± 1.6	8.03 ± 1.6
	AP Avg	5.91 ± 1.4	7.66 ± 1.4	7.57 ± 1	7.37 ± 1	7.23 ± 1.5	6.92 ± 1.3	6.99 ± 1.64
L3/4	Anterior	8.37 ± 2.3	10.22 ± 1.5	10.19 ± 1.5	9.76 ± 1.1	9.73 ± 1.8	9.13 ± 1.9	9.69 ± 1.6
	Middle	9.63 ± 2.3	10.12 ± 1.4	10.36 ± 1	12.1 ± 1.7	10.51 ± 2.1	10.12 ± 1.9	10.95 ± 2.2
	Posterior	5.7 ± 1.4	6.14 ± 1.5	6.32 ± 0.9	6.84 ± 1	5.62 ± 2	5.61 ± 1.1	5.88 ± 1.9
	Average	7.62 ± 1.8	8.82 ± 1.1	8.96 ± 0.9	9.57 ± 1	8.62 ± 1.8	8.29 ± 1.4	8.81 ± 1.8
	AP Avg	6.67 ± 1.8	8.18 ± 1.3	8.26 ± 0.9	8.3 ± 0.9	7.67 ± 1.7	7.37 ± 1.3	7.76 ± 1.7
L4/5	Anterior	10.22 ± 2.3	12.03 ± 1.6	11.39 ± 2.2	10.97 ± 1.4	10.8 ± 2.2	10.35 ± 2.3	10.52 ± 2.4
	Middle	10.06 ± 1.6	10.51 ± 1.5	10.63 ± 1.3	11.79 ± 1.4	10.81 ± 2.2	10.54 ± 2	10.38 ± 1.7
	Posterior	5.92 ± 1.2	6.37 ± 1.6	6.78 ± 1.2	6.88 ± 1.1	5.6 ± 1.5	5.67 ± 1.3	5.54 ± 1.5
	Average	8.73 ± 1.5	9.64 ± 1.2	9.6 ± 1.4	9.9 ± 1.1	9.07 ± 1.7	8.85 ± 1.7	8.78 ± 1.7
	AP Avg	8.7 ± 1.6	9.2 ± 1.4	9.09 ± 1.5	8.92 ± 1.1	8.21 ± 1.6	8.01 ± 1.7	7.98 ± 1.8
L5/S1	Anterior	10.24 ± 2.4	12.96 ± 2.3	12.01 ± 2.9	11.7 ± 3.2	12.07 ± 2.2	11.21 ± 2.4	12.42 ± 3
	Middle	9.18 ± 2.3	9.25 ± 1.2	9.26 ± 2.2	10.35 ± 2.2	9.09 ± 2.7	9.17 ± 2.1	9.81 ± 2.2
	Posterior	5.09 ± 1.1	5.23 ± 1.2	5.35 ± 1.3	5.36 ± 1.3	5.59 ± 1.8	5.08 ± 1.3	5 ± 1.5
	Average	8.17 ± 1.6	9.15 ± 1.1	8.88 ± 1.9	9.14 ± 1.8	8.91 ± 1.6	8.49 ± 1.6	9.08 ± 2.1
	AP Avg	7.67 ± 1.52	9.1 ± 1.2	8.68 ± 1.9	8.53 ± 1.8	8.83 ± 1.5	8.71 ± 2.1	8.71 ± 2.1

AP Avg, average of only anterior and posterior disc height.

DISCUSSION

To our knowledge, the present study represents the first and most extensive of its kind in accurately establishing normal values for individual lumbar intervertebral disc height. At present, there is a widely held belief that disc height declines with age. Our results did not demonstrate this trend. It appears that in healthy individuals, disc height has more limited and less straightforward variation with age. Specifically, our results suggest that for upper lumbar levels, middle disc height increases, whereas anterior and posterior heights remain unchanged. Conversely, for lower lumbar levels (except L5/S1), middle disc height remains unchanged, whereas anterior and posterior heights diminish. The discs seem to undergo a process of elliptization, that is, progression from a rectangular cross section to an elliptical one, confirming findings of previous, albeit more limited, studies.[8,10,12,13,16] Figure 9 graphically demonstrates the elliptization process.

Part of the importance of restoration of disc height is indirect decompression of neural elements by restoring FH. It is noteworthy that FH decline with age reached stronger correlation, and more frequently reached statistical significance, than either overall

Table 3. Female Disc Height by Decade of Life

Level	Location	First Decade	Second Decade	Third Decade	Fourth Decade	Fifth Decade	Sixth Decade	Seventh Decade
T12/L1	Anterior	4.46 ± 0.7	5.26 ± 0.9	5.18 ± 1.2	4.38 ± 0.9	4.56 ± 1.3	4.85 ± 1	4.88 ± 1
	Middle	6.02 ± 1.5	5.93 ± 0.9	5.92 ± 0.8	6.07 ± 1.1	6.3 ± 1	6.4 ± 1	6.31 ± 1.2
	Posterior	3.54 ± 1	3.36 ± 0.7	3.45 ± 1	2.94 ± 0.8	3.3 ± 0.8	3.32 ± 0.9	3.37 ± 0.8
	Average	4.68 ± 0.9	4.85 ± 0.5	4.84 ± 0.8	4.46 ± 0.7	4.72 ± 0.8	4.86 ± 0.8	4.9 ± 0.9
	AP Avg	4 ± 0.72	4.31 ± 0.6	4.3 ± 0.9	3.66 ± 0.7	3.93 ± 0.9	4.09 ± 0.85	4.13 ± 0.8
L1/2	Anterior	5.86 ± 0.9	6.21 ± 1.1	6.41 ± 1.2	5.22 ± 1.4	5.93 ± 1.2	6.17 ± 1.3	5.65 ± 1.3
	Middle	6.65 ± 1.1	6.84 ± 0.9	7 ± 0.9	7.24 ± 1.3	7.51 ± 1.2	7.64 ± 1.2	7.11 ± 1.9
	Posterior	4.7 ± 0.7	4.46 ± 1	4.57 ± 0.9	4.12 ± 0.7	4.35 ± 0.7	4.16 ± 0.9	4 ± 1.4
	Average	5.74 ± 0.8	5.84 ± 0.7	5.99 ± 0.8	5.53 ± 0.9	5.93 ± 0.8	5.99 ± 1	5.81 ± 0.9
	AP Avg	5.28 ± 0.7	5.34 ± 0.9	5.49 ± 0.9	4.67 ± 0.9	5.14 ± 0.8	5.16 ± 1	4.8 ± 1.3
L2/3	Anterior	7.29 ± 1.3	7.44 ± 1.4	7.53 ± 1.4	6.87 ± 1.5	6.82 ± 1.2	7.35 ± 1.6	6.75 ± 2.2
	Middle	8.28 ± 1.9	8.42 ± 1	8.2 ± 1.3	8.67 ± 1.3	8.64 ± 1.3	9.09 ± 1.7	8.09 ± 2.2
	Posterior	5.24 ± 1	5.57 ± 1	5.45 ± 0.9	5.29 ± 0.9	4.96 ± 0.9	4.77 ± 1.2	4.52 ± 1.1
	Average	6.94 ± 1	7.14 ± 0.9	7.06 ± 1	6.94 ± 0.9	6.81 ± 0.9	7.07 ± 1.3	6.45 ± 1.8
	AP Avg	6.27 ± 0.9	6.5 ± 1	6.49 ± 1	6.08 ± 0.9	5.89 ± 0.9	6.06 ± 1.2	5.63 ± 1.6
L3/4	Anterior	7.94 ± 1.1	9.01 ± 1.8	9.14 ± 1.6	7.5 ± 1.4	7.68 ± 1.3	7.66 ± 1.8	7.63 ± 1.1
	Middle	9.44 ± 2.1	9.26 ± 1.4	9.12 ± 1.5	8.97 ± 1.2	9.4 ± 1.4	9.57 ± 2.1	8.37 ± 1.6
	Posterior	5.88 ± 1.6	6.2 ± 1	6.14 ± 1.2	5.78 ± 1.1	5.23 ± 1.2	5.18 ± 1	4.56 ± 1
	Average	7.75 ± 1.4	8.16 ± 1.1	8.13 ± 1.2	7.42 ± 0.9	7.43 ± 1	7.47 ± 1.2	6.85 ± 1.2
	AP Avg	6.91 ± 1.22	7.61 ± 1.1	7.64 ± 1.3	6.64 ± 1	6.45 ± 1	6.42 ± 1.2	6.09 ± 1
L4/5	Anterior	9.83 ± 1.1	11.62 ± 1.9	11.33 ± 1.8	9.06 ± 1.6	8.97 ± 1.8	8.4 ± 2.5	9.1 ± 2.5
	Middle	9.64 ± 1.9	9.95 ± 1.4	9.99 ± 1.3	9.54 ± 1.4	9.74 ± 1.9	9.22 ± 2.3	9.69 ± 2.6
	Posterior	6.02 ± 1.6	6.39 ± 1.3	6.71 ± 1.3	6.13 ± 1.3	5.51 ± 1.1	5.55 ± 1.6	5.46 ± 1.6
	Average	8.5 ± 1.1	9.32 ± 1.1	9.34 ± 1.2	8.24 ± 1.1	8.07 ± 1.4	7.72 ± 2	8.11 ± 2.1
	AP Avg	7.93 ± 1	9.01 ± 1.2	9.02 ± 1.3	7.6 ± 1.2	7.24 ± 1.3	6.97 ± 2	7.32 ± 2
L5/S1	Anterior	10.73 ± 2.2	13.16 ± 2.5	13.8 ± 2.4	10.32 ± 2.1	10.8 ± 3	10.41 ± 2	9.35 ± 3.5
	Middle	9.35 ± 2.2	9.17 ± 1.4	9.75 ± 1.6	8.86 ± 2.2	9 ± 2.4	9.17 ± 1.6	8.76 ± 2.7
	Posterior	4.79 ± 1.5	5.24 ± 1.5	5.7 ± 0.9	5.21 ± 1.6	5.12 ± 1.4	5.33 ± 1.4	5.02 ± 1.7
	Average	8.29 ± 1.4	9.19 ± 1.3	9.76 ± 1.4	8.13 ± 1.7	8.32 ± 2.1	8.30 ± 1.5	7.71 ± 2.5
	AP Avg	7.76 ± 1.5	9.2 ± 1.4	9.77 ± 1.4	7.76 ± 1.6	7.96 ± 2	7.87 ± 1.5	7.18 ± 2.4

AP Avg, average of only anterior and posterior disc height.

average or AP average disc height, despite our limited sample size. This would suggest that bony remodeling of the vertebral bodies themselves, both increasing concavity of the endplate and overall loss of height, are prominent features of the aging process of the spine, even in the absence of clinical pathology. Vertebral body endplate deflection (inward budging), accompanied by deformation of the trabecular bone, have been demonstrated as part of the response to axial loading of the spine.[17,18] Damage to the trabecular bone can occur even at small strain such as the repetitive strain of habitual loading rather than trauma.[19] As the normally ageing nucleus pulposus stiffens, and vertebral bodies lose mass and density, the aforementioned factors likely contribute to the morphologic changes in vertebral bodies and accompanying loss of height.

It is not difficult to imagine that the typical surgeon's clinical experience of near ubiquitous degenerative changes that occur in intervertebral discs with age could give rise to the perception of uncomplicated disc height decline with age. However, as our

Table 4. Male Foraminal Height by Decade of Life

Level	Location	First Decade	Second Decade	Third Decade	Fourth Decade	Fifth Decade	Sixth Decade	Seventh Decade
T12/L1	Average	16.59 ± 1.9	16.42 ± 1.3	16.43 ± 1.2	16.78 ± 2.3	15.94 ± 1.6	16.17 ± 1.5	16.2 ± 2
	Left	16.61 ± 1.8	16.15 ± 1.4	16.33 ± 1.3	16.53 ± 2.3	15.83 ± 1.7	15.97 ± 1.6	16.03 ± 2.1
	Right	16.57 ± 2.1	16.69 ± 1.4	16.35 ± 1.2	17.03 ± 2.2	16.06 ± 1.6	16.38 ± 1.8	16.36 ± 2
L1/2	Average	18.73 ± 1.6	18.69 ± 1.4	18.39 ± 1.4	18.94 ± 1.5	18.03 ± 1.5	18.08 ± 1.6	18.13 ± 1.7
	Left	18.52 ± 1.8	18.84 ± 1.4	18.26 ± 1.5	18.69 ± 1.6	17.91 ± 1.	18 ± 1.5	18.31 ± 1.8
	Right	18.95 ± 1.6	18.53 ± 1.7	18.51 ± 1.4	19.18 ± 1.5	18.14 ± 1.6	18.16 ± 1.8	17.95 ± 1.7
L2/3	Average	19.99 ± 1.5	19.67 ± 1.5	19.85 ± 1.4	20.78 ± 1.3	19.21 ± 1.7	18.84 ± 1.9	18.92 ± 1.7
	Left	20 ± 1.8	19.57 ± 1.6	19.71 ± 1.7	20.95 ± 1.6	19.08 ± 1.8	18.51 ± 2	18.39 ± 2.1
	Right	19.99 ± 1.3	19.78 ± 1.6	19.98 ± 1.3	20.62 ± 1.2	19.34 ± 1.7	19.17 ± 2	19.45 ± 1.8
L3/4	Average	19.74 ± 1.7	19.13 ± 1.7	18.9 ± 1.3	19.77 ± 1.4	18.42 ± 1.7	17.62 ± 2	17.78 ± 1.7
	Left	19.83 ± 1.9	19.16 ± 1.8	18.8 ± 1.2	20.02 ± 1.5	18.34 ± 1.6	17.75 ± 2.7	17.65 ± 2
	Right	19.65 ± 1.7	19.09 ± 1.9	19 ± 1.5	19.53 ± 1.5	18.5 ± 1.9	17.49 ± 2.3	17.91 ± 1.6
L4/5	Average	18.77 ± 1.4	18.53 ± 1.4	18.18 ± 1.4	18.39 ± 1.2	17.48 ± 2	16.69 ± 2	16.39 ± 2
	Left	19.08 ± 1.5	18.62 ± 1.6	18.45 ± 1.56	19.13 ± 1.4	17.64 ± 2	16.83 ± 2.1	17.10 ± 2
	Right	18.45 ± 1.3	18.44 ± 1.5	17.91 ± 1.7	17.66 ± 1.4	17.31 ± 2.2	16.55 ± 2.3	15.68 ± 2.6
L5/S1	Average	15.27 ± 0.7	15.61 ± 1.6	14.66 ± 1.6	14.79 ± 1.6	13.96 ± 2	12.73 ± 2.2	14.28 ± 2
	Left	15.35 ± 0.8	15.60 ± 1.7	15.02 ± 2.1	14.47 ± 1.3	13.77 ± 2.3	12.63 ± 2.5	14.05 ± 1.9
	Right	15.20 ± 1.2	15.61 ± 1.7	14.31 ± 1.5	15.12 ± 2.1	14.15 ± 2.2	12.84 ± 2.5	14.50 ± 2.7

Table 5. Female Foraminal Height by Decade of Life

Level	Location	First Decade	Second Decade	Third Decade	Fourth Decade	Fifth Decade	Sixth Decade	Seventh Decade
T12/L1	Average	15.38 ± 1.7	16.18 ± 1.8	15.53 ± 2	15.3 ± 1.6	15.45 ± 1.84	14.68 ± 1.9	14.35 ± 1.84
	Left	15.3 ± 2.1	16.07 ± 1.7	15.32 ± 1.9	15.17 ± 1.9	15.3 ± 2	14.47 ± 2.1	14.10 ± 2.1
	Right	15.46 ± 1.6	16.3 ± 2	15.74 ± 2.1	15.42 ± 1.5	15.61 ± 1.8	14.88 ± 1.9	14.61 ± 1.9
L1/2	Average	18.36 ± 2.3	17.78 ± 1.7	17.68 ± 2	17.46 ± 1.4	17.43 ± 1.6	16.6 ± 1.8	15.7 ± 2.1
	Left	18.18 ± 2.4	17.76 ± 1.7	17.79 ± 2.2	17.39 ± 1.5	17.32 ± 1.9	16.73 ± 2.2	15.61 ± 2
	Right	18.55 ± 2.3	17.8 ± 1.9	17.57 ± 1.8	17.52 ± 1.4	17.54 ± 1.5	16.47 ± 1.7	15.79 ± 2.4
L2/3	Average	19.61 ± 2.4	19.46 ± 1.7	18.83 ± 1.8	19.04 ± 1.5	18.77 ± 1.5	17.82 ± 2.5	16.57 ± 1.85
	Left	19.66 ± 2.4	19.57 ± 1.6	18.82 ± 1.9	19.03 ± 1.7	18.78 ± 1.6	18.05 ± 2.4	16.48 ± 1.8
	Right	19.56 ± 2.7	19.35 ± 2	18.84 ± 2	19.05 ± 1.5	18.75 ± 1.4	17.6 ± 3	16.65 ± 2.4
L3/4	Average	18.89 ± 2.43	18.86 ± 1.6	18.63 ± 2.1	18.49 ± 1.6	17.97 ± 1.44	17.26 ± 2.4	15.49 ± 2.3
	Left	19.27 ± 2.4	18.97 ± 1.7	18.75 ± 2	18.47 ± 1.7	18 ± 1.8	17.2 ± 2.4	15.85 ± 2.6
	Right	18.52 ± 2.6	18.75 ± 1.6	18.5 ± 2.2	18.5 ± 1.6	17.94 ± 1.4	17.31 ± 2.5	15.12 ± 2.2
L4/5	Average	17.63 ± 2.1	18.33 ± 1.5	18.17 ± 1.6	17.87 ± 1.7	17.58 ± 1.3	16.22 ± 2.3	15.37 ± 2.8
	Left	17.48 ± 2.2	18.59 ± 1.9	18.32 ± 1.5	17.88 ± 1.8	17.56 ± 1.64	16.40 ± 2.5	15.64 ± 2.7
	Right	17.42 ± 2.1	18.07 ± 1.5	18.02 ± 1.7	17.86 ± 1.9	17.60 ± 1.3	16.04 ± 2.5	15.09 ± 3.1
L5/S1	Average	13.88 ± 1.6	14.94 ± 1.8	15.26 ± 1.3	14.65 ± 1.8	13.89 ± 2.1	13.85 ± 1.9	13.51 ± 1.1
	Left	13.74 ± 1.4	14.77 ± 2.1	15.1 ± 1.4	14.47 ± 2	13.85 ± 2.4	13.44 ± 2	13.23 ± 1.53
	Right	14.02 ± 1.9	15.1 ± 1.8	15.42 ± 1.5	14.84 ± 1.6	13.94 ± 2	14.25 ± 2.3	13.78 ± 1.6

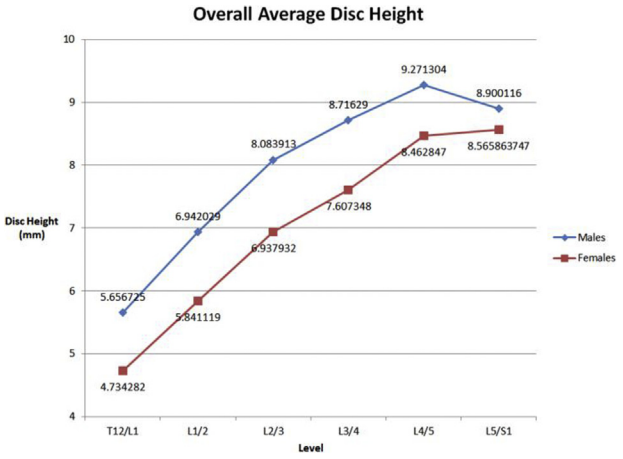


Figure 8. Variation of overall average disc height by sex and by level.

Table 6. Independent *t*-test for Disc and Foraminal Height

	Men (N = 109)		Women (N = 131)		
Measurement	**Mean**	**SD**	**Mean**	**SD**	***P* Value**
T12/L1 Foraminal height average	16.27	1.56	15.35	1.86	0.000
T12/L1 Anterior disc height	6.29	1.49	4.81	1.09	0.000
T12/L1 Middle disc height	6.66	1.34	6.13	1.04	0.001
T12/L1 Posterior disc height	3.75	1.00	3.29	0.82	0.000
T12/L1 Average disc height	5.02	1.13	4.05	0.81	0.000
L1/2 Foraminal height average	18.34	1.50	17.29	1.87	0.000
L1/2 Anterior disc height	7.59	1.58	5.92	1.27	0.000
L1/2 Middle disc height	8.20	1.82	7.18	1.22	0.000
L1/2 Posterior disc height	4.96	1.27	4.31	0.92	0.000
L1/2 Average disc height	6.28	1.26	5.12	0.95	0.000
L2/3 Foraminal height average	19.46	1.66	18.63	2.02	0.001
L2/3 Anterior disc height	8.70	1.67	7.13	1.51	0.000
L2/3 Middle disc height	9.79	1.87	8.51	1.46	0.000
L2/3 Posterior disc height	5.67	1.45	5.14	1.04	0.001
L2/3 Average disc height	7.18	1.39	6.14	1.10	0.000
L3/4 Foraminal height average	18.58	1.81	18.00	2.12	0.025
L3/4 Anterior disc height	9.66	1.75	8.11	1.62	0.000
L3/4 Middle disc height	10.41	1.86	9.16	1.54	0.000
L3/4 Posterior disc height	5.90	1.50	5.59	1.22	0.079
L3/4 Average disc height	7.78	1.44	6.85	1.21	0.000
L4/5 Foraminal height average	17.65	1.91	17.43	2.06	0.387
L4/5 Anterior disc height	10.94	2.15	9.82	2.28	0.000
L4/5 Middle disc height	10.64	1.76	9.70	1.79	0.000
L4/5 Posterior disc height	6.05	1.42	5.98	1.41	0.705
L4/5 Average disc height	8.50	1.61	7.90	1.62	0.005
L5/S1 Foraminal height average	14.30	2.05	14.36	1.80	0.806
L5/S1 Anterior disc height	11.92	2.59	11.36	2.94	0.123
L5/S1 Middle disc height	9.34	2.12	9.13	2.00	0.444
L5/S1 Posterior disc height	5.27	1.38	5.23	1.43	0.851
L5/S1 Average disc height	8.59	1.65	8.30	1.87	0.198

SD, standard deviation.

results demonstrate, there is no clear straightforward correlation of a change in disc height with age in healthy individuals, and therefore this perception represents the conflation of 2 distinct but not necessarily independent processes—aging and degeneration.

Furthermore, our data demonstrated a strong correlation between disc height and sex. Intervertebral discs in women were significantly smaller than in men at all levels except for L5/S1. It is unclear what the underlying cause of this sexual dimorphism in disc height stems from, but it may be related to BMI or muscle mass. Indeed, both height and weight were significantly larger in men than in women; however, it is unclear that these impact disc height in any significant way.

Surprisingly, very little literature exists on normal values of intervertebral disc height, given the context that its restoration is one of the main goals of fusion with interbody spacers. The lack of a clear target for disc height is further compounded by a scarcity of literature on selection of implant size. We found only 1 study that noted the average cage height used at their institution, which ranged from 11.25 ± 1.32 mm to 12.38 ± 1.43 mm.[20] The largest average intervertebral disc height observed in our study was 9.87 ± 0.7 mm for men in their eighth decade of life, notably less than the smallest implants used in the study. Although it is unclear what rationale the average spine surgeon uses in cage height selection, caution should be taken to avoid the use of oversized cages, as the immediate consequence of doing so is an increase in pressure at the interface between the implant and vertebral endplates, potentially resulting in loss of regional lordosis.[1-4]

The most obvious complication that increased vertebral endplate pressure predisposes to is endplate subsidence, but can include failure of the procedure and need for reoperation. The problem of endplate subsidence and its relationship to other complications—such as loss of correction—and clinical consequences—such as return of pain—is poorly understood. Foremost, there is no unified grading or valuating system for subsidence.[5-7] Aside from excessive cage height, several other factors contribute to subsidence including more anterior versus posterior cage placement, cage width, and cage lordosis.[5-7] Thus,

Table 7. Male Age Correlation of Disc and Foraminal Height

	Anterior Disc Height		Middle Disc Height		Posterior Disc Height		Average Disc Height		Anterior-Posterior Average		Average Foraminal Height	
Disc Level	R	*P* Value	R	*P* Value	R	*P* Value	R	*P* Value	R	*P* Value	R	*P* Value
T12/L1	0.087	0.369	0.403*	0.000	0.193†	0.044	0.265*	0.005	0.143	0.137	−0.107	0.267
L1/2	0.012	0.899	0.341*	0.000	0.155	0.107	0.211†	0.027	0.086	0.376	−0.181	0.060
L2/3	−0.066	0.493	0.258*	0.007	−0.020	0.836	0.080	0.406	−0.050	0.602	−0.247*	0.010
L3/4	−0.074	0.446	0.124	0.201	−0.059	0.541	0.003	0.976	−0.076	0.435	−0.355*	0.000
L4/5	−0.202†	0.035	0.016	0.867	−0.246†	0.010	−0.163	0.091	−0.243†	0.010	−0.430*	0.000
L5/S1	−0.053	0.588	0.012	0.904	−0.040	0.677	−0.034	0.726	−0.058	0.547	−0.415*	0.000

Correlative strength was determined as R = 0.1—0.39 as weak, 0.4—0.59 as moderate and >0.6 as strong.
*Correlation is significant at the 0.01 level (2-tailed).
†Correlation is significant at the 0.05 level (2-tailed).

it is difficult to ascertain to what extent the seemingly widespread phenomenon of oversized cage selection contributes to suboptimal outcomes for patients. However, based on our data and currently available literature, the use of cages ranging from 6—10 mm—appropriately selected based on patient's sex and overall body height, combined with mid-vertebral body placement, maximum possible cage width, and the use of lordotic rather than square profile cages—would be a prudent approach for restoring both disc height and lumbar lordosis, while minimizing the risk of endplate subsidence and other associated complications. Future study prospectively analyzing correlation between restored disc height and cage subsidence, while controlling for lordosis or cage location, is needed to estimate its contribution to this phenomenon.

The current study is limited in several ways. Foremost is the relatively small number of patients. Although the total number of 240 patients may seem adequate, when divided between 2 sexes and 7 decades, that is 14 groups; it may be insufficient, and thus, have obfuscated trends that would have been apparent with a larger patient number. Although the trends in our analysis did not always reach statistical significance, statistical and clinical significance are not always equivalent. Finally, although we did attempt to determine any potential effects of height, weight, or BMI, these data were not available for all subjects (81 women, 69 men). Nonetheless, to our knowledge, the current study is the largest of its kind. Further subgrouping would shrink the groupings beyond usefulness; however, the aforementioned factors remain a critical area for future study. Our measurements were

Table 8. Female Age Correlation of Disc and Foraminal Height

	Average Disc Height		Middle Disc Height		Posterior Disc Height		Average Disc Height		Anterior-Posterior Average		Average Foraminal Height	
Disc Level	R	*P* Value	R	*P* Value	R	*P* Value	R	*P* Value	R	*P* Value	R	*P* Value
T12/L1	−0.091	0.302	0.161	0.067	−0.035	0.691	0.017	0.847	−0.079	0.368	−0.275*	0.001
L1/2	−0.063	0.472	0.195†	0.025	−0.186†	0.033	−0.004	0.960	−0.132	0.134	−0.345*	0.000
L2/3	−0.105	0.232	0.060	0.496	−0.315*	0.000	−0.122	0.167	−0.221†	0.011	−0.413*	0.000
L3/4	−0.265*	0.002	−0.074	0.399	−0.424*	0.000	−0.304*	0.000	−0.390*	0.000	−0.447*	0.000
L4/5	−0.387*	0.000	−0.067	0.449	−0.248*	0.004	−0.291*	0.001	−0.380*	0.000	−0.412*	0.000
L5/S1	−0.388*	0.000	−0.085	0.333	−0.041	0.643	−0.252*	0.004	−0.320*	0.000	−0.241*	0.006

Correlative strength was determined as R = 0.1—0.39 as weak, 0.4—0.59 as moderate and >0.6 as strong.
*Correlation is significant at the 0.01 level (2-tailed).
†Correlation is significant at the 0.05 level (2-tailed).

Table 9. Women: Height, Weight, and Body Mass Index

Age (Years)	Height (cm)	Weight (kg)	BMI (kg/m^2)
18	162.6	61.7	23.3
18	162.6	59	22.2
19	161	81.3	31.3
19	162.6	50.8	19.1
19	170.2	65.3	22.6
19	170.2	72.3	24.9
20	168.9	81.6	28.6
21	167.6	85.5	30.4
21	167.6	86.2	30.7
22	165.1	83	30.5
23	160	94	36.7
23	172.7	102.1	34.2
24	180.3	91	30.5
26	170.2	92.5	32
27	185.4	72.6	21.1
27	172.7	76.7	25.7
27	177.8	104.3	33
28	157.5	73	29.6
30	160	62.3	24.3
32	166	82.1	29.8
33	170.2	87.5	30.3
33	167.6	75.8	27
34	172.7	104.3	35
36	152.4	60.8	26.2
36	165	61.7	22.7
36	171.2	75.7	26.2
38	160	71.7	28
39	160	51.7	20.2
39	160	68.2	33.7
39	157.5	52.6	21.2
40	162.6	60.8	23
40	162.6	107.5	40.5
41	162.6	56.2	21.2
41	156.2	97.8	21.6
42	157.5	107	43.2
42	162.6	69.4	26.3
42	167.6	92.1	32.8
42	162	90.5	34.5
44	162.6	60.3	22.8
45	172.7	105	35.2
45	152.4	56.7	24.4

Continues

Table 9. Continued

Age (Years)	Height (cm)	Weight (kg)	BMI (kg/m^2)
48	160	123.4	48.2
48	157.5	60.3	24.3
48	172.7	51.3	17.2
49	160	77.1	30.1
49	157.5	71.5	28.8
50	167.6	80.3	28.6
51	157.5	55.4	22.3
53	167.6	69.4	24.7
54	162.6	73.5	27.8
54	177.8	120.7	38.2
55	167.6	112.5	40
55	152.4	82.1	35.3
56	167.6	73.3	26
58	167.6	80.6	28.7
58	165.1	91.2	33.4
58	157.5	80.6	33.3
59	165.1	86.8	31.8
59	160	50	19.5
59	157.5	71.8	31.5
60	166.4	67.6	24.5
61	163.8	59.9	27.7
62	162.6	50	18.9
63	157.5	90.7	36.6
64	165.1	99.6	36.5
65	160	59.8	23.4
65	171.5	66.7	22.7
66	160	55.8	21.8
66	157.5	64	25.8
67	163.8	71.2	26.5
69	152.4	90.2	38.8
72	165.1	71.7	26.3
72	162.6	78.2	29.6
72	154.9	65.8	27.4
74	149.9	78.9	35.1
74	157.5	48.1	19.4
76	170.2	73.5	25.4
76	167.6	73.9	26.3
77	154.9	50.3	21
77	149.9	64.8	28.9
77	172.7	85.3	28.6

BMI, body mass index.

Table 10. Men: Height, Weight, and Body Mass Index

Age (Years)	Height (cm)	Weight (kg)	BMI (kg/m^2)
18	169.7	88.1	27.7
18	177.8	82.6	26.1
19	180.3	124.6	38.2
19	175.3	68.9	22.5
19	170.2	95.3	33
20	170.2	68.5	23.7
21	172.7	71.7	24
22	185.4	81.2	23.6
22	193	131.5	35.8
22	177.8	90.7	28.7
22	182.9	88.5	26.4
23	177.8	70.3	26.4
24	175.3	86.2	28.1
28	184.2	98.4	29
28	185.4	120.2	35
31	177.8	56.2	17.8
32	177.8	120.2	38
33	193	138.5	37.2
33	177.8	98.9	31.3
35	172.7	71.2	23.9
35	180.3	103	31.7
36	182.9	116.1	34.7
37	185.4	91.6	26.7
38	180.3	110.7	34.2
38	175.3	98	32
39	177.8	84.8	26.8
40	182.9	57.5	17.2
40	175.3	77.8	25.3
40	177.8	104.1	32.9
40	177.8	91.6	29
40	191.8	98	26.6
49	175.3	68	22.2
50	185.4	85.7	21.8
50	177.8	99.8	31.6
51	185.4	104.3	30.3
51	180.3	64.4	19.8
52	180.3	132.5	40.7
52	177.8	70	22.1
57	195.6	117.5	30.7
58	180.3	66.7	20.5
59	185.4	100.9	29.3
59	185.4	123.6	36
60	185.4	91	26.5
61	170.2	80.3	27.7
62	185.4	108.9	31.7
63	167.6	71.7	23.7
64	180.3	87.8	27
64	177.8	92.1	29.1
65	182.9	82.6	24.7
65	175.3	85.3	27.8
65	177.8	92.1	29.1
66	182.9	79.4	23.7
66	167.6	75.3	26.7
66	182.9	90.7	27.1
67	170.2	61.2	21.1
67	172.7	87.8	29.4
68	167.6	83.9	29.9
68	195.6	88	23
68	182.9	93	33.9
69	177.8	104.3	32.1
69	167.6	88.9	31.6
69	182.9	93	27.8
72	188	143.3	40.6
73	173	70.9	23.7
77	175.3	62.6	20.4
78	177.8	68.5	21.7
78	175.3	62.6	20.4
79	167.6	77.1	27.4
80	175.3	66.7	21.8

BMI, body mass index.

made using CT data, and we acknowledge the disc heights may be different using data from magnetic resonance imaging in healthy individuals. The recumbent position, likewise, may have contributed to a positional change in height. However, we expect the trend per each level and each decade of life should be the same.

CONCLUSIONS

Contrary to widespread belief, variation in disc height is determined much more by sex than by age. Although not unequivocally demonstrated, there appears to be a process of elliptization of the

Table 11. Comparison of Men and Women: Age, Height, Weight, and Body Mass Index

	Age	Height	Weight	BMI
Women (n = 81)	46.5	163.8	76.2	28.4
SD	17.8	6.93	17.5	6.13
Men (n = 69)	48.5	179.1	90	27.9
SD	19.3	6.71	20.2	5.4
Mean Diff	−2.04	−15.34	−13.81	0.42
P value	0.502	**<0.0005**	**<0.0005**	0.66

Bold values indicate statistical significance ($P < 0.05$).
BMI, body mass index; SD, standard deviation; Diff, difference.

Table 12. Men and Women: Correlation of Height, Weight, and Body Mass Index to Age

Correlation of Age to	Height	Weight	BMI
Women (n = 81)	−0.332	−0.139	−0.003
P value	**0.002**	0.217	0.976
Men (n = 69)	−0.083	−1.67	−0.158
P value	0.499	0.171	0.196

Bold value indicates statistical significance ($P < 0.05$).
BMI, body mass index.

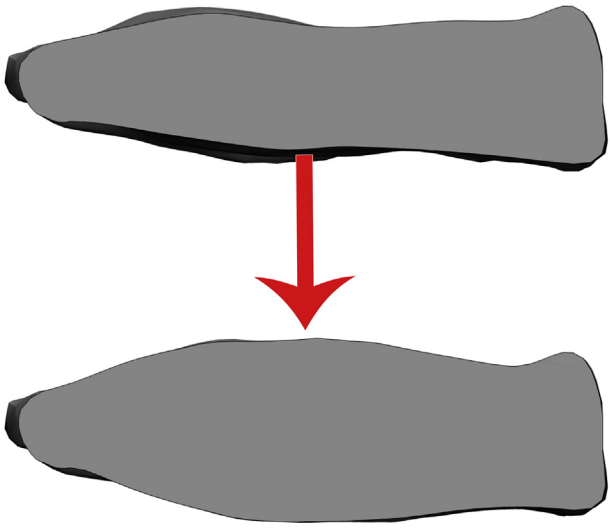


Figure 9. Graphical illustration of elliptization undergone by the aging human intervertebral disc.

cross section of intervertebral disc. The maximum height of the interbody space in the adult lumbar spine was at the L4/5 level (8.9 ± 1.7 mm for men; 8.6 ± 1.8 mm for women). Measured disc heights were smaller for both men and women than typically selected cage sizes, as reported in the literature. The extent to which this contributes to endplate subsidence, related complications, and clinical outcomes remains to be elucidated, as does the greater picture of the aging spine, including remodeling and decline of the vertebra. The extent these factors are impacted by height, weight, and BMI remains to be more fully elucidated as well.

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